

Conditionally Approved: A Proof-Bound Mortgage Benchmark for Long-Horizon AI Agents Under Changing Evidence, Authority, and Time

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Pre-results working draft - 24 August 2026

Contents

Abstract	2
1. Why conditional approval is the right evaluation problem	2
2. Research questions	2
3. Related work and measurement boundary	3
4. Benchmark design	3
5. Evaluation contract	4
6. Locked analysis plan	5
7. Results	5
8. Validity and limitations	6
9. Ethics, rights, and release	6
10. Reproducibility	6
11. Relationship to the method paper	6
12. Counterfactual research program beyond this panel	6
References	7
Appendix A. Frozen identity registry	7
Appendix B. Human-review decision ledger	7

Pre-results boundary. This manuscript names the frozen study and locks its reporting structure while execution is in progress. It does not yet claim model performance, comparative superiority, frontier difficulty, human agreement, mortgage-domain validity, causal effects, or novelty. Every empirical statement must be regenerated from the sealed machine-readable panel and must survive the evaluator- and human-admission gates described below.

Abstract

Mortgage work is not a static question-answering problem. A loan file evolves while people and systems act: documents arrive incomplete, evidence expires or is retracted, policies are superseded, deadlines advance, authorities disagree, and work crosses shifts. We introduce **Conditionally Approved**, a fully synthetic, proof-bound benchmark for evaluating long-horizon AI agents under those changes. Five task editions isolate document truth under pushback, epistemic residue after retraction, competing legitimate authority, temporal portfolio control, and an honest handoff whose quality is measured by successor performance.

The benchmark binds every eligible episode to a frozen task edition, model identity, runtime commit, action ledger, material-world roots, agent-visible projections, event and restore lineage, exact evaluator edition, and infrastructure disposition. Official reward is a strict binary conjunction over task correctness and evidence obligations; diagnostic vectors are reported separately and never converted into unregistered partial reward. We initially preregistered five attempts for each of four identity-pinned model providers. Before completing that design, the operator imposed a prospective cost stop: a hard ceiling of 55 physical attempts and a balanced target of two gradable attempts per provider $\times$ task cell. The amendment preserves every prior attempt, selects the lowest two gradable ordinals within each exact cell, excludes infrastructure failures from behavioral denominators, and forbids pass-at-five or tail-reliability claims.

Exact-cell pass-at-two, strict two-attempt consistency, cost, trajectory shape, failure signatures, and Analytics+ interpretations will be inserted only after the cost-capped inventory is sealed and independently verified. Missing cells remain missing rather than being replaced by a pooled statistic. Behavioral interpretation is reserved for blinded human review. The benchmark was built and verified with the Gradia Universes platform; Gradia is the execution and evidence system, not the benchmark name.

1. Why conditional approval is the right evaluation problem

An agent can produce a plausible final answer and still fail the work. It may cite a superseded source, silently obey an unauthorized request, miss a deadline boundary, carry retracted evidence into later decisions, or hand off a summary that causes the next agent to violate an active constraint. Conversely, a long or inefficient trajectory is not automatically incorrect. Conditioned on valid infrastructure, the official score asks whether the submitted work is correct and fully evidence-grounded under the frozen contract—not whether the agent used a preferred number of steps.

The title has a deliberate double meaning. In mortgage operations, approval is often conditional on unresolved evidence. In this benchmark, every evaluation claim is also conditional on a valid world, an admitted evaluator, and reviewable evidence.

2. Research questions

- **RQ1 — Exact completion.** Does an identity-pinned frontier agent produce a perfect-rubric completion in each observed evolving workflow?
- **RQ2 — Two-attempt repeatability.** In exact provider $\times$ task cells with two same-runtime gradable attempts, does the model pass at least once, and does it pass both times? This is exploratory repeatability evidence, not a tail-reliability or general capability estimate.
- **RQ3 — Behavioral differentiation.** Which evidence, authority, temporal, and handoff obligations produce distinct, repeatable failure signatures?
- **RQ4 — Measurement validity.** Do deterministic checks, calibrated model-based interpretation, and blinded humans agree on which failures are attributable to the agent rather than the environment or evaluator?
- **RQ5 — Operational cost.** What calls, tokens, tool actions, wall time, and provider spend are required for a valid attempt, and how much missingness is caused by infrastructure rather than behavior?

3. Related work and measurement boundary

Dynamic and stateful agent benchmarks already establish that useful evaluation must go beyond static question answering. Gaia2/Agents Research Environments studies asynchronous changing worlds; InterruptBench studies user revisions and retractions; and tau-bench, tau2-bench, AppWorld, ToolSandbox, and OSWorld evaluate agents through actions that alter persistent environments [1–7]. Those systems occupy the broad territory of dynamic environments, interruptions, and stateful tool use. This work does not claim those ideas as new.

The narrower measurement problem is whether a benchmark can prove the chain from a declared occurrence to the material world transition, the exact projection visible to the agent, the eligible action boundary, any restore lineage, and the final judgment. Proof of Execution, CAVA, agent-native telemetry, W3C PROV, and recent runtime-evidence work establish adjacent execution, provenance, canonical-action, and telemetry foundations [8–14]. Agentic Data Environments and BranchBench make the complementary systems case that consequential agents operate over a closure of files, databases, processes, applications, APIs, memory, and derived state, while Data Flow Control studies semantic source-to-sink restrictions [15–17]. The method paper therefore tests a composition and its incremental detection value; it does not present hashing, logging, branching, provenance, or checkpointing as novel in isolation.

Benchmark-quality work also warns that a clean headline score can conceal broken tasks, evaluator leakage, contamination, or a saturated ceiling [18–20]. Conditionally Approved responds with mutation-tested evaluators, immutable preregistration, explicit infrastructure sensors, runtime-stratified denominators, and human adjudication. These controls make the result more auditable; they do not substitute for domain-expert validity or a sufficiently powered behavioral study.

4. Benchmark design

4.1 Frozen five-task panel

Task edition	Core stressor	Required capability	Registered treatment
Document truth under pushback	Incomplete and conflicting document evidence	Reconcile exact pages, versions, claims, and current authority	<code>alder.page-missing</code>
Epistemic residue	Evidence observed, retracted, and replaced	Stop using dead evidence and re-ground the decision	<code>cedar.seen-retracted-replaced</code>
Authority and fair judgment	Two legitimate principals conflict	Apply authority scope, protect fairness constraints, and escalate correctly	<code>dogwood.competing-legitimate-principals</code>
Temporal portfolio control	Coupled files evolve around a commitment boundary	Coordinate scarce capacity, deadlines, and post-change rechecks	<code>fir.phase-pre-commit</code>
Honest handoff and verified commit	Work crosses an agent/session boundary	Preserve active constraints, uncertainty, provenance, and successor usability	<code>birch.contract-altered</code>

The stories, people, organizations, values, documents, and policies are synthetic. They are not customer records, mortgage advice, production policy, or evidence of external-domain validity.

4.2 Long-horizon execution

The safe scripted controls require 113–150 actions under full load. That observation establishes instrument solvability and the existence of a long execution path; it does not establish frontier difficulty. Model attempts may use fewer or more actions. Act count, verbosity, latency, and cost are diagnostic variables rather than reward criteria unless a task explicitly binds a deadline or resource limit.

4.3 Evolving-world evidence

Each declared occurrence binds its frozen event identity, exact eligible action boundary, before/after material roots, agent-visible projection, authority, restore generation, and previous occurrence-chain head. This lets the verifier distinguish environment application, disclosure, authority, restore, evaluator, and agent faults before an episode enters a model-performance denominator.

4.4 Formal episode and disposition contract

An episode is represented as

$$E_i = (T_i, M_i, S_i, W_{i,0}, A_i, \Omega_i, R_i, J_i),$$

where T_i is the frozen task edition, M_i the requested and resolved model identity, S_i the scaffold and runtime identity, $W_{i,0}$ the initial material world root, A_i the ordered act ledger, Ω_i the occurrence-witness chain, R_i the restore lineage, and J_i the evaluator edition and criterion vector. Every occurrence $\omega_{i,k}$ binds the declared event, eligible boundary, before/after roots, visible projection, authority, restore generation, and the previous occurrence digest. A verifier recomputes these bindings from released bytes; it does not trust a reported scalar.

Before scoring, every physical attempt receives exactly one disposition:

1. **gradable** — protocol, identity, runtime, world, and evaluator evidence close;
2. **infrastructure censor** — dispatch, transport, provider, or host evidence makes behavior unscorable;
3. **measurement defect** — the environment or evaluator cannot support the intended attribution; or
4. **pending adjudication** — deterministic observations exist, but their cause has not yet been assigned.

Only the first disposition may enter a behavioral denominator. The others remain visible, retain cost and receipt evidence, and cannot be silently retried into a preferred outcome.

5. Evaluation contract

5.1 Official reward

Official reward is 1 only when every registered binary requirement passes. It is 0 otherwise. Criterion vectors, branch probes, tool-use traces, and narrative analyses are diagnostic evidence—not hidden partial credit. A correct and complete result is not penalized merely for taking more actions.

Formally, if $g_{i,1}, \dots, g_{i,K_i} \in \{0, 1\}$ are the registered applicable gates, the official episode outcome is

$$y_i = \prod_{k=1}^{K_i} g_{i,k}.$$

The conjunction prevents a strong final narrative from compensating for stale evidence, missing authority, an unauthorized irreversible act, or an unusable handoff. Separately reported criterion coverage can help diagnose where an episode failed, but it is not partial credit and cannot change y_i .

5.2 Evaluator admission

The evaluator package must demonstrate:

1. a positive control for every task edition;
2. isolated one-defect mutations for each criterion;
3. current-source citations backed by root-owned read evidence;
4. post-boundary re-reading of every materially changed resource;
5. exact task, runtime, world-root, and evaluator-edition binding;
6. blinded independent human review of the criteria and representative outcomes; and
7. a fail-closed disposition for unresolved measurement defects.

Deterministic red criteria are initially **unresolved observations**. They become model-attributable failure modes only after adjudication rules exclude environment, provider, and evaluator defects.

5.3 Panel, amendment, and denominator

The original plan crossed four identity-pinned provider/model cells with five task editions and five gradable attempts per cell. It was not completed. The cost-cap amendment instead defines 20 exact provider $\times$ task cells, targets two gradable attempts per cell, and limits the entire execution to 55 physical attempts. Infrastructure failures retain their receipts and costs but do not enter behavioral denominators. Within a cell, the lowest two

gradable ordinals are selected; later gradable attempts remain disclosed as overage and cannot replace an earlier result. An exact-cell pair is reportable only if both selected attempts bind the same task, evaluator, model identity, Git commit, and runtime-implementation digest.

The execution wrapper was allowed to remain active while the repository advanced. The post-run verifier therefore partitions attempts by exact runtime cohort and refuses pooled panel claims if more than one cohort is present. A later code gate latches the panel commit before dispatch and rejects any child that drifts. The historical cohort split is retained as a measurement defect, not edited away.

6. Locked analysis plan

For each provider $\times$ task cell, report:

- physical attempts, gradable attempts, and infrastructure exclusions;
- the exact runtime cohort of every selected attempt;
- **pass@2**: whether at least one of the two selected same-runtime attempts succeeds, only where an eligible pair exists;
- **pass²**: whether both selected same-runtime attempts succeed, only where an eligible pair exists;
- single-attempt outcomes where exactly one gradable attempt exists, without relabeling them as pass@2;
- infrastructure exclusions and reasons;
- calls, input/output tokens, tool actions, wall time, and cost;
- registered criterion-failure frequencies, initially labeled unresolved; and
- identity-blinded Analytics+ interpretations with cited trajectory evidence and human disposition.

Across the panel, report inventory summaries without pooling away task, provider, or runtime identity. A balanced statistic exists only if all 20 exact cells have two eligible attempts under one common runtime cohort. No significance test, causal claim, provider ranking, pass@5 claim, difficulty claim, or novelty claim may be added after seeing outcomes unless it belongs to a separately preregistered study.

For an eligible exact cell c with the two selected binary outcomes $(y_{c,1}, y_{c,2})$, this draft uses two deliberately different descriptive quantities:

$$\text{pass@2}_c = \max(y_{c,1}, y_{c,2}), \quad \text{pass}^2_c = y_{c,1}y_{c,2}.$$

The first asks whether either observed attempt passed; the second asks whether both did. With only two attempts, neither is a stable tail-reliability or population capability estimate. No value is emitted if the two attempts differ in frozen task, model, evaluator, Git commit, or runtime-implementation digest.

7. Results

Locked shell — do not hand-edit empirical values. Tables and figures in this section must be generated from the sealed aggregate artifact.

7.1 Completion and exclusions

Pending sealed panel.

7.2 Exact-cell outcomes and repeatability

Pending sealed panel.

7.3 Cost and trajectory shape

Pending sealed panel.

7.4 Criterion-level observations

Pending blinded human adjudication.

7.5 Analytics+ findings

Pending evidence-cited interpretation and human disposition.

8. Validity and limitations

Five synthetic tasks and a cost-capped, runtime-stratified panel cannot establish mortgage-domain validity, broad agent capability, tail reliability, training utility, or real-world safety. A wall of zero scores may identify a floor, an over-constrained evaluator, or both; it does not by itself establish that the tasks are valuable. A wall of perfect scores may identify a ceiling; it does not justify making tasks arbitrarily obscure. Follow-on editions should be admitted from observed failure evidence, expert review, and controlled anchor tiers rather than post-hoc difficulty chasing.

The registered treatment cells do not alone identify causal effects of retraction, phase, authority, or handoff. Those claims require exact seed-paired controls or counterfactual branches, repeated executions, closure evidence, and a separately admitted analysis.

9. Ethics, rights, and release

The public benchmark assets are synthetic. Model outputs, raw trajectories, derived publication, redistribution, and training use are separate rights surfaces and must be reviewed per artifact and provider. Public release requires the exact signed edition, source and evaluator identities, redaction decision, independent review, and a machine-verifiable release receipt. No customer or partner attribution is permitted without written authorization tied to the exact released bytes.

10. Reproducibility

The release will include the frozen task and evaluator editions, model and runtime identities, preregistrations, provider-normalized receipts, aggregate generator, verification commands, result tables, figure sources, and a rendered PDF. Any raw artifact withheld for rights or security reasons must be named in a reconstruction- gap report rather than silently omitted.

11. Relationship to the method paper

The companion manuscript, *Interruptible Universes: Evolution Witnesses for Verifiable World Change in Agent Benchmarks*, defines and tests the general evidence composition. **Conditionally Approved** is the benchmark and empirical study built on that apparatus. The separation is intentional: the method should be judged by its validity evidence, and the benchmark should be judged by task quality, measurement quality, and results.

12. Counterfactual research program beyond this panel

This five-task panel is a first measurement, not the endpoint. The same frozen-world apparatus supports a sequence of stronger studies, each of which needs its own preregistration and control conditions:

- **Epistemic residue:** compare a world in which evidence was seen and later retracted with a seed-paired world in which it was never introduced. The post-retraction behavioral delta is descriptive until repeated forks, isolation, and human validation support a causal interpretation.
- **Interruption phase-response:** apply the same authoritative event at different semantic action boundaries and estimate phase-specific response curves without changing any other world fact.
- **Authority ladders:** vary authenticated identity, channel, scope, urgency, and conflicts between legitimate principals while preserving exact message content.
- **Honest handoff:** end agent A's session, expose agent B only to the frozen handoff artifact, and score A partly through B's ability to honor active constraints and changed evidence.
- **Act-level counterfactual credit:** fork a valid whole-world snapshot before one observed act, substitute an admitted reference continuation, and report the terminal utility delta. Reference-policy recoverability must remain separate from causal blame and training-grade advantage labels.
- **Branch-consistency oversight:** ask an agent for an evidence-cited account, then rerun concealed counterfactual forks that alter one witnessed fact. This can test account consistency, but it cannot establish deception or truthfulness without independent ground truth, anti-gaming controls, calibrated humans, and acceptable false-accusation rates on honest controls.
- **Gauge-invariant evaluation:** transform only non-semantic names, IDs, serialization order, timestamp rendering, and causally incomparable event order. A criterion that flips under a valid re-projection reveals

presentation dependence rather than a world-state distinction.

The engineering repository contains pre-results contracts or fixtures for portions of this program. Their existence is not behavioral evidence. The next scientific step is to show that the world and evaluator detect controlled defects with low false rejection before using branches to draw conclusions about live agents.

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Appendix A. Frozen identity registry

Generated after the panel seals; includes task, plan, runtime, model, evaluator, artifact, aggregate, and release digests.

Appendix B. Human-review decision ledger

Generated from blinded review; records agreement, disagreement, adjudication, and any task/evaluator defect without rewriting historical model outcomes.